

Enrichment Videos: Parallelograms

Day 11 Enrichment · MathCraft · Grade 6 · Professor Xavier Honablué, M.Ed.

ENRICHMENT

How to use this page: Start with the **6th-grade** videos. The **Advanced Outlook** videos go beyond 6th-grade standards — they explain the "det A" and " λ " labels from the Stretching Machine lesson (the Warp Gate story). Pause often and answer the Watch-For questions in your math notebook. MIT, Stanford, and Carnegie Mellon sources are marked.

Level 1 · 6th-Grade Standard (6.G.A.1)

KHAN ACADEMY	Khan Academy — Intuition for area of a parallelogram Watch for: where the triangle is cut off and where it slides. Why does the area stay the same?
MATH ANTICS	Math Antics — Areas of Parallelograms (YouTube) Watch for: which measurement is the height, and why the slanted side is NOT used.
STANFORD	YouCubed at Stanford — video channel Stanford Graduate School of Education videos on visual math and a growth mindset. Watch for: one way to "see" a math idea with a picture.

Level 2 · Advanced Outlook (beyond 6th grade)

3BLUE1BROWN	3Blue1Brown — The determinant (Essence of Linear Algebra, Ch. 6) Animated video by Grant Sanderson, a Stanford math graduate. It shows a unit square stretched into a parallelogram — our exact picture. Watch for: what "det" measures, and what happens to every other shape.
MIT	MIT OpenCourseWare — Lecture 18: Properties of Determinants (Prof. Gilbert Strang) A real MIT college lecture. Watch just the first 5–10 minutes. Watch for: the word "determinant" and how a professor writes a matrix.
MIT	MIT BLOSSOMS — Math & Science Video Lessons Free MIT video lessons for high-school classes. Browse the math lessons for a geometry topic that interests you.
CARNEGIE MELLON	CMU CS Academy (Carnegie Mellon University) Not a video series — we could not find a free Carnegie Mellon parallelogram video. Instead, this free CMU course lets you code shapes with (x, y) points. Try drawing the Leaning Constellation.

Watch-For Questions (answer in your notebook)

1. In your own words: why is the area of a parallelogram $\text{base} \times \text{height}$ and not $\text{base} \times \text{slanted side}$?
2. Draw a parallelogram and mark its height with a dotted line and a small right-angle box.
3. Advanced Outlook: In the 3Blue1Brown video, what does it mean when the determinant is 3? Connect it to Chapter 5 of the Warp Gate story.
4. Advanced Outlook: In the Warp Gate story, the Golden Lane was never bent. Find a moment in a video where a line stays on its own path while being stretched.
5. What is one question you still have? Write it down and bring it to class.

Families: These are free, public videos. Links open outside our site; please preview YouTube videos with your student. Advanced Outlook videos are optional and will not be graded.